



Time Series Analysis

Seminar 8: Long memory and ARFIMA



Daniel Traian PELE

Bucharest University of Economic Studies

IDA Institute Digital Assets

Blockchain Research Center

AI4EFin Artificial Intelligence for Energy Finance

Romanian Academy, Institute for Economic Forecasting

MSCA Digital Finance

Acronyms used in this material

- ▣ **ACF**: Autocorrelation Function
- ▣ **AI**: Artificial Intelligence
- ▣ **AR**: AutoRegressive (model)
- ▣ **ARCH**: AutoRegressive Conditional Heteroskedasticity
- ▣ **ARFIMA**: AutoRegressive Fractionally Integrated Moving Average (model)
- ▣ **ARIMA**: AutoRegressive Integrated Moving Average (model)
- ▣ **ARMA**: AutoRegressive Moving Average (model)
- ▣ **BET**: Bucharest Exchange Trading (index)
- ▣ **BIC**: Bayesian Information Criterion
- ▣ **BNR**: Banca Națională a României (the National Bank of Romania)
- ▣ **CPI**: Consumer Price Index
- ▣ **DFA**: Detrended Fluctuation Analysis
- ▣ **EODHD**: EOD Historical Data (market-data provider)
- ▣ **FIGARCH**: Fractionally Integrated GARCH
- ▣ **FRED**: Federal Reserve Economic Data
- ▣ **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ **GPH**: Geweke–Porter–Hudak (estimator)
- ▣ **HICP**: Harmonised Index of Consumer Prices
- ▣ **LW**: Local Whittle (estimator)
- ▣ **MA**: Moving Average
- ▣ **ML**: Machine Learning
- ▣ **OLS**: Ordinary Least Squares
- ▣ **RV**: Realised Volatility
- ▣ **SE**: Standard Error
- ▣ **US**: United States

Today's Question and Seminar Structure

- The question of the seminar
 - ▶ **does a shock to a series fade at an exponential rate (ARMA) or much more slowly (long memory), and how do we measure the difference?**
- Results to obtain (practice only: nothing is handed in or graded)
 - ▶ the answers to the A exercises
 - ▶ a figure and a table for each B exercise
 - ▶ the corrected AI answer of [C2](#)
 - ▶ one sentence of interpretation for each exercise
- Notebook for today: [open the seminar notebook in Google Colab](#)
 - ▶ each task names its notebook section
- The exercises, their types and their order: [the exercise map](#) (next slide)
- Seminar: prof. Daniel Traian Pele, danpele@ase.ro

Exercise map

Task	Question	Type	Model
A1, A2	weights of $(1 - L)^d$, ACF of ARFIMA(0, d , 0), ranges of d	Solved, Proposed	A1
A3, A4	R/S and H by hand; a GPH slope by hand	Solved, Proposed	A3
A5, A6	an ARFIMA forecast; GARCH and FIGARCH weights	Solved, Proposed	A5
B1, B2	the Nile and its break; Romanian inflation	Solved, Proposed	B1
B3, B4	S&P 500 volatility; BET and EUR/RON	Solved, Proposed	B3
C1, C2	US inflation and regimes; what is wrong in an AI answer?	Proposed	B1, B2

- Types: **A** calculations on paper; **B** real data, in the notebook; **C** open questions, an AI critique
- **[Solved]**: full solution in the slides and in the notebook, a model to follow; **[Proposed]**: you solve it, following the model; the solution is discussed in the seminar

Data used

Series	Source	Frequency	Period
Nile flow at Aswan	statsmodels (nile)	yearly	1871–1970
HICP, Romania	Eurostat (prc_hicp_minr)	monthly, 2015 = 100	2005–2026
CPI, United States	FRED (CPIAUCSL)	monthly, SA	1947–2026
S&P 500, BET	EODHD	daily, close	2000–2026
EUR/RON	BNR reference rate	daily	2005–2026

- Romanian inflation: monthly rate $100\Delta \ln P_t$ minus its monthly means (seasonal adjustment), and the 12-month rate $100(\ln P_t - \ln P_{t-12})$
- In the notebook: `gph`, `local_whittle`, `hurst_rs`, `arfima_exact_ml`, `monthly_rv`; no account or key is needed

Prerequisites for Today (1/22): autocorrelation and memory

- ▣ **Autocovariance** and **autocorrelation** of a stationary series x_t at lag k

$$\gamma(k) = \text{Cov}(x_t, x_{t-k}), \quad \rho(k) = \frac{\gamma(k)}{\gamma(0)}$$

- ▶ $\gamma(0) = \text{Var}(x_t)$; $\rho(k) \in [-1, 1]$: the correlation between two values k periods apart

- ▣ **Sample autocorrelation** (the ACF of the data, [Chapter 1](#))

$$\hat{\rho}(k) = \frac{\sum_{t=k+1}^T (x_t - \bar{x})(x_{t-k} - \bar{x})}{\sum_{t=1}^T (x_t - \bar{x})^2}$$

- ▶ T : the number of observations; $\bar{x}$: the sample mean; band of no correlation: $\pm 1.96/\sqrt{T}$

- ▣ **Memory**: how fast $\rho(k)$ goes to zero as the lag k grows

- ▶ fast decay: a shock is forgotten quickly; slow decay: the distant past still matters

Prerequisites for Today (2/22): short memory and long memory

- ▣ **Short memory** (ARMA models, [Chapter 2](#)): exponential decay

$$|\rho(k)| \leq C r^k, \quad 0 < r < 1$$

- ▶ $C > 0$: a constant; r : the decay factor per lag; AR(1): $\rho(k) = \phi^k$, so $r = |\phi|$
- ▶ the autocorrelations are summable: $\sum_k |\rho(k)| < \infty$

- ▣ **Long memory**: hyperbolic (power-law) decay

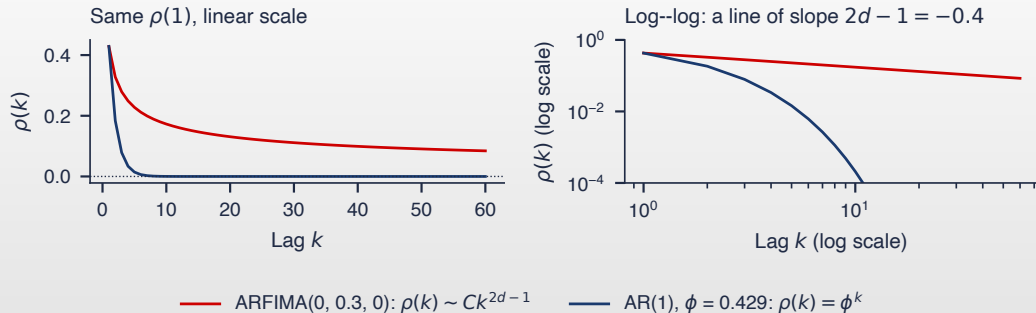
$$\rho(k) \sim C k^{2d-1}, \quad 0 < d < \frac{1}{2}$$

- ▶ d : the **memory parameter**; $\sim$: the ratio of the two sides tends to 1 as $k \rightarrow \infty$
- ▶ the exponent $2d - 1$ lies in $(-1, 0)$: the decay is so slow that $\sum_k \rho(k) = \infty$

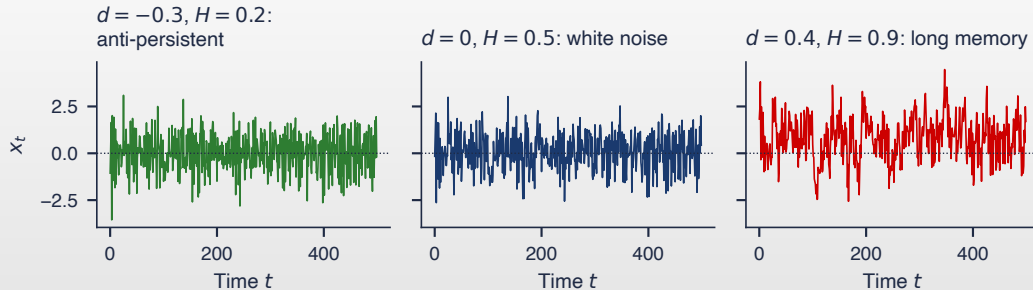
- ▣ **Hurst exponent** $H = d + 1/2$

- ▶ $H = 0.5$: no memory; $H > 0.5$: persistence; $H < 0.5$: anti-persistence
- ▶ Hurst found $H \approx 0.7$ for the yearly flow of the Nile [Hurst \(1951\)](#)

Prerequisites for Today (3/22): the two decays in a chart



- Left: ARFIMA(0, 0.3, 0) and an AR(1) with the same $\rho(1) = 0.429$
 - ▶ at lag 10: $\rho(10) = 0.173$ for ARFIMA, but only $0.429^{10} = 0.0002$ for the AR(1)
- Right: the same curves on log-log axes
 - ▶ $\log \rho(k) \approx \log C + (2d - 1) \log k$: a power law is a straight line of slope $2d - 1 = -0.4$
 - ▶ the exponential decay bends down and leaves the chart after about 10 lags

Prerequisites for Today (4/22): simulated paths for three values of d 

- ▣ The same shocks ε_t (white noise, variance 1), filtered with three values of d (the model is defined on the next slides)
- ▣ $d = -0.3$ ($H = 0.2$): **anti-persistent**
 - ▶ a value above the mean tends to be followed by one below: $\rho(1) < 0$, the path looks rougher than white noise
- ▣ $d = 0.4$ ($H = 0.9$): **persistent**
 - ▶ long excursions above or below the mean, which look like trends or changes of level, although the mean is constant

Prerequisites for Today (5/22): the lag operator and fractional differencing

- **Lag operator** L : $Lx_t = x_{t-1}$, $L^k x_t = x_{t-k}$
 - ▶ first difference: $(1 - L)x_t = x_t - x_{t-1}$, the case $d = 1$
- **Fractional difference** $(1 - L)^d$ for a real d : the binomial series

$$(1 - L)^d x_t = \sum_{k=0}^{\infty} \pi_k x_{t-k}, \quad \pi_0 = 1, \quad \pi_k = \pi_{k-1} \frac{k-1-d}{k}$$

- ▶ π_k : the weight of the value x_{t-k} ; $d = 1$ gives $\pi_1 = -1$ and $\pi_k = 0$ for $k \geq 2$ (the usual difference)
 - ▶ for $0 < d < 1$: every $\pi_k < 0$ ($k \geq 1$) and $|\pi_k|$ decreases slowly, like k^{-1-d} : all past values count
- Example, $d = 0.4$: $\pi_1 = -0.4$, $\pi_2 = -0.4 \cdot 0.6/2 = -0.12$, $\pi_3 = -0.12 \cdot 1.6/3 = -0.064$

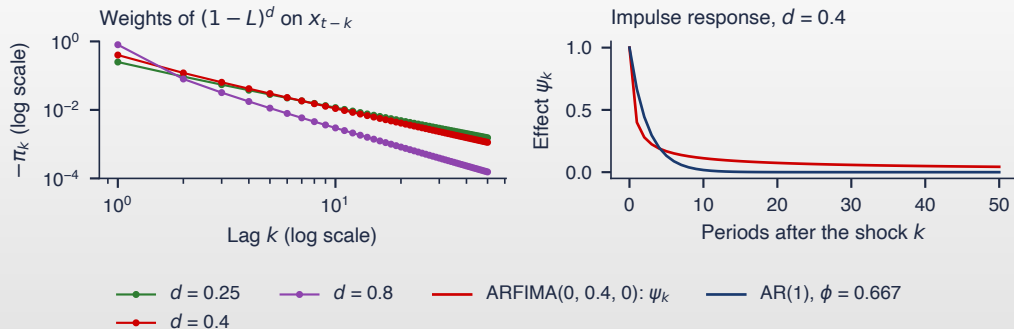
Prerequisites for Today (6/22): the impulse response ψ_k

- ▣ **Inverse filter:** the series as a sum of past shocks (the $MA(\infty)$ form)

$$x_t = (1 - L)^{-d} \varepsilon_t = \sum_{k=0}^{\infty} \psi_k \varepsilon_{t-k}, \quad \psi_0 = 1, \quad \psi_k = \psi_{k-1} \frac{k - 1 + d}{k}$$

- ▶ ψ_k : the effect on x_t of a unit shock that happened k periods ago
- ▣ Example, $d = 0.4$: $\psi_1 = 0.4$, $\psi_2 = 0.4 \cdot 1.4/2 = 0.28$, $\psi_3 = 0.28 \cdot 2.4/3 = 0.224$
 - ▶ for $0 < d < 1/2$, ψ_k decreases like k^{d-1} : slowly, but towards 0, so the series is stationary
- ▣ For an $AR(1)$, $\psi_k = \phi^k$: the effect of a shock dies out exponentially

Prerequisites for Today (7/22): fractional weights and impulse responses in a chart



- Left: the weights $-\pi_k$ of $(1 - L)^d$ on log-log axes are almost straight lines: a power law in k
- Right: for $d = 0.4$, $\psi_{10} = 0.112$, against $0.667^{10} = 0.017$ for an AR(1) with the same $\rho(1) = 0.667$

Prerequisites for Today (8/22): the ARFIMA model

- **ARFIMA**(p, d, q): an ARMA model for the fractionally differenced series [Granger and Joyeux \(1980\)](#), [Hosking \(1981\)](#)

$$\phi(L)(1-L)^d(x_t - \mu) = \theta(L)\varepsilon_t$$

- ▶ $\phi(L) = 1 - \phi_1 L - \dots - \phi_p L^p$, $\theta(L) = 1 + \theta_1 L + \dots + \theta_q L^q$: the AR and MA polynomials ([Chapter 2](#))
- ▶ μ : the mean; ε_t : white noise with variance σ^2 ; $d = 0$: ARMA; $d = 1$: ARIMA($p, 1, q$)

Range of d	Properties
$d \leq -1/2$	not invertible
$-1/2 < d < 0$	stationary, invertible, anti-persistent
$d = 0$	short memory (ARMA)
$0 < d < 1/2$	stationary, invertible, long memory
$1/2 \leq d < 1$	not stationary, but mean-reverting: a shock still dies out
$d = 1$	unit root: a shock is permanent

- For $1/2 \leq d < 3/2$, difference once: $(1-L)x_t$ is ARFIMA with memory parameter $d - 1$

Prerequisites for Today (9/22): ARFIMA(0, d , 0) in numbers

▣ **Autocorrelations** by recursion

$$\rho(1) = \frac{d}{1-d}, \quad \rho(k) = \rho(k-1) \frac{k-1+d}{k-d}, \quad k \geq 2$$

- ▶ example $d = 0.3$: $\rho(1) = 0.3/0.7 = 0.429$, $\rho(2) = 0.429 \cdot 1.3/1.7 = 0.328$
- ▶ for large k : $\rho(k) \approx \frac{\Gamma(1-d)}{\Gamma(d)} k^{2d-1}$, the constant C of the long-memory definition

▣ **Variance**

$$\gamma(0) = \sigma^2 \frac{\Gamma(1-2d)}{\Gamma(1-d)^2}$$

- ▶ $\Gamma(z) = \int_0^\infty u^{z-1} e^{-u} du$: the gamma function; $\Gamma(n) = (n-1)!$ for integers $n \geq 1$; $\Gamma(1/2) = \sqrt{\pi}$
- ▶ $d > 0$: $\gamma(0) > \sigma^2$, the memory accumulates past shocks; $\gamma(0) \rightarrow \infty$ as $d \rightarrow 1/2$

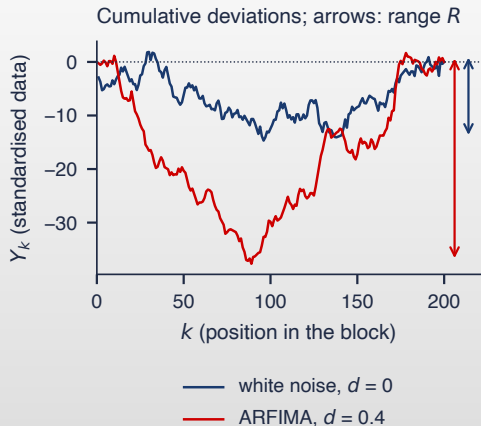
Prerequisites for Today (10/22): the one-step forecast

- From $(1 - L)^d(x_{T+1} - \mu) = \varepsilon_{T+1}$, replace the unknown shock ε_{T+1} by its mean, 0

$$\hat{x}_{T+1} = \mu - \sum_{k \geq 1} \pi_k (x_{T+1-k} - \mu)$$

- ▶ x_T : the last observation; x_{T+1-k} : the value $k - 1$ periods before it; in practice the sum stops at $k = T$
- Reading the formula for $0 < d < 1$
 - ▶ the weights $-\pi_k$ are positive: each past deviation from μ pushes the forecast in its own direction
 - ▶ the weights decrease slowly, so old deviations still count
- Comparison: AR(1), $\hat{x}_{T+1} = \mu + \phi(x_T - \mu)$
 - ▶ only the last value enters the forecast; ϕ : the AR(1) coefficient

Prerequisites for Today (11/22): the R/S statistic



- For a block $x_1, \dots, x_n$ with mean $\bar{x}$ [Hurst \(1951\)](#):

$$Y_k = \sum_{i=1}^k (x_i - \bar{x}), \quad k = 1, \dots, n$$

- ▶ Y_k : the cumulative deviation from the mean; $Y_n = 0$

- **Range and standard deviation**

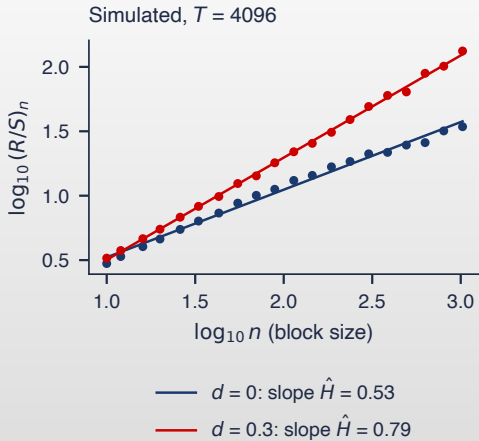
$$R = \max_k Y_k - \min_k Y_k, \quad S = \sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2}$$

- ▶ R/S : the range of the cumulative deviations, in units of S ; it grows like n^H

- Chart: a persistent block has long runs of the same sign

- ▶ Y_k drifts far from 0, so R is large (39 against 17 for white noise)

Prerequisites for Today (12/22): H from a log-log regression



- Split the series in blocks of size n and average R/S over the blocks: $(R/S)_n$

$$\log_{10}(R/S)_n \approx a + H \log_{10} n$$

- ▶ repeat for several n ; $\hat{H}$ = the OLS slope of $\log_{10}(R/S)_n$ on $\log_{10} n$; $\hat{d} = \hat{H} - 0.5$
- ▶ a : the intercept; OLS: ordinary least squares
- Chart: white noise gives $\hat{H} = 0.53$, slightly above 0.5
- ▶ R/S is biased upwards in small blocks; $d = 0.3$ gives $\hat{H} = 0.79 \approx 0.8$
- DFA (detrended fluctuation analysis) [Peng et al. \(1994\)](#)
 - ▶ cumulate the series, fit a line in each block of size n ; $F(n)$: the root mean square of the residuals
 - ▶ $F(n) \propto n^H$: $\hat{H}$ = the slope of $\log F(n)$ on $\log n$

Prerequisites for Today (13/22): the periodogram and the spectral density

- ▣ **Fourier frequencies** $\lambda_j = 2\pi j/T$, $j = 1, \dots, \lfloor T/2 \rfloor$
 - ▶ λ_j in radians per period; it corresponds to a cycle of T/j periods; $\lfloor \cdot \rfloor$: the integer part
 - ▶ low frequencies (small j): long cycles; this is where long memory shows
- ▣ **Periodogram**: how much of the variance is in cycles of frequency λ_j

$$I(\lambda_j) = \frac{1}{2\pi T} \left| \sum_{t=1}^T (x_t - \bar{x}) e^{-i\lambda_j t} \right|^2$$

- ▶ i : the imaginary unit; $I(\lambda_j)$ is a noisy estimate of the spectral density $f(\lambda_j)$
- ▣ **Long memory** = a pole of the spectral density at frequency zero

$$f(\lambda) \sim G \lambda^{-2d} \quad \text{as } \lambda \rightarrow 0$$

- ▶ $G > 0$: a constant; ARFIMA(0, d , 0): $f(\lambda) = \frac{\sigma^2}{2\pi} (4 \sin^2(\lambda/2))^{-d}$, and $4 \sin^2(\lambda/2) \approx \lambda^2$ for small λ

Prerequisites for Today (14/22): the GPH estimator

- **GPH Geweke and Porter-Hudak (1983)**: take logs of $I(\lambda_j) \approx f(\lambda_j)$ at the m lowest frequencies: a linear regression in d

$$\log I(\lambda_j) = c + d X_j + u_j, \quad X_j = -\log(4 \sin^2(\lambda_j/2)), \quad j = 1, \dots, m$$

- ▶ c : the intercept; u_j : the error; m : the **bandwidth**, the number of low frequencies used

- The estimate is the OLS slope, with $y_j = \log I(\lambda_j)$

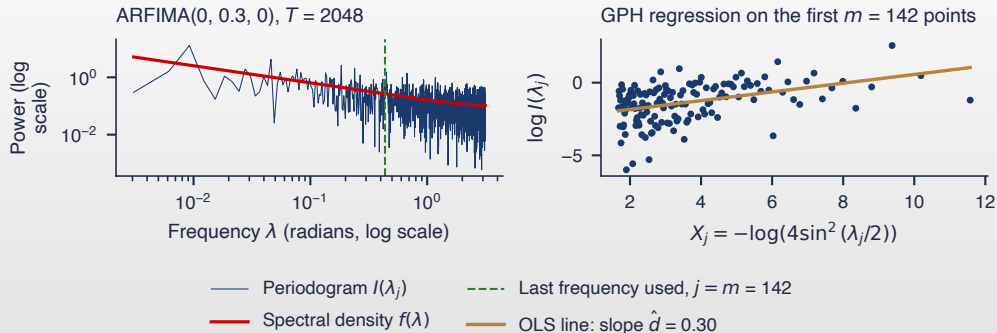
$$\hat{d} = \frac{\sum_{j=1}^m (X_j - \bar{X})(y_j - \bar{y})}{\sum_{j=1}^m (X_j - \bar{X})^2}, \quad \text{SE}(\hat{d}) = \frac{\pi}{\sqrt{24m}}$$

- ▶ $\bar{X}$, $\bar{y}$: the means over the m frequencies; SE: the asymptotic standard error

- **Test of H_0 : $d = 0$** (no long memory)

- ▶ $t = \hat{d}/\text{SE}(\hat{d})$, compared with the Normal distribution: reject at 5% if $|t| > 1.96$
- ▶ usual bandwidth $m = \lfloor T^{0.65} \rfloor$; a small m : less bias from short-run dynamics, but more noise

Prerequisites for Today (15/22): the periodogram and GPH in a chart



- Left: the periodogram of a simulated ARFIMA(0, 0.3, 0) ($T = 2048$) scatters around $f(\lambda)$, which rises towards $\lambda = 0$
- Right: the GPH regression on the first $m = \lfloor 2048^{0.65} \rfloor = 142$ frequencies gives $\hat{d} = 0.30$ ($\text{SE} = \pi / \sqrt{24 \cdot 142} = 0.054$)

Prerequisites for Today (16/22): local Whittle, exact ML and the information criteria

- **Local Whittle (LW)** [Robinson \(1995\)](#): $\hat{d}$ minimises, over the same m frequencies,

$$R(d) = \log \left(\frac{1}{m} \sum_{j=1}^m \lambda_j^{2d} I(\lambda_j) \right) - \frac{2d}{m} \sum_{j=1}^m \log \lambda_j$$

- ▶ a likelihood that uses only $f(\lambda) \approx G\lambda^{-2d}$ near zero; $\text{SE}(\hat{d}) = 1/(2\sqrt{m})$, smaller than for GPH

- **Exact maximum likelihood (ML)** [Sowell \(1992\)](#)

- ▶ maximises the full Normal likelihood L of ARFIMA(p, d, q) over d , the ϕ_i , the θ_j and σ^2
- ▶ uses all frequencies: more precise if the model is right, biased if the short-run part is wrong

- **Information criteria**: the model with the smallest value wins

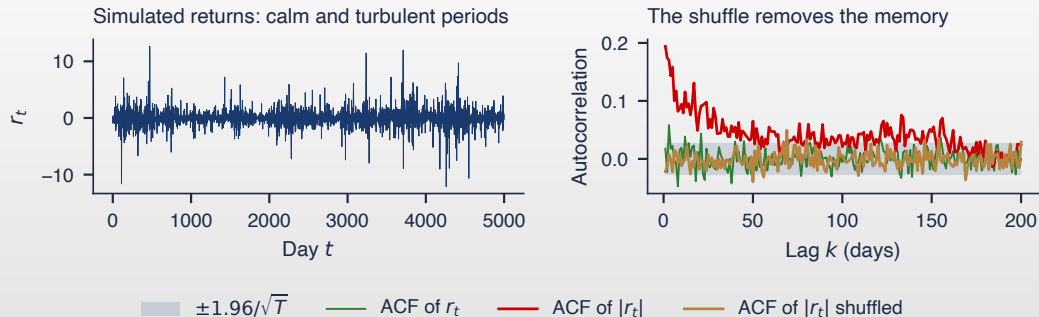
$$\text{AIC} = -2 \ln L + 2k, \quad \text{BIC} = -2 \ln L + k \ln T$$

- ▶ In L : the maximised log-likelihood; k : the number of estimated parameters; BIC penalises extra parameters more

Prerequisites for Today (17/22): memory in volatility and the shuffle test

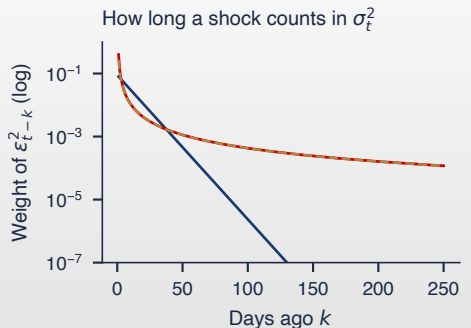
- Returns $r_t = 100 \ln(P_t/P_{t-1})$ are almost uncorrelated, but their size $|r_t|$ has long memory [Ding, Granger and Engle \(1993\)](#)
 - ▶ P_t : the closing price of day t ; monthly **realised volatility** $\sqrt{RV_m}$, with $RV_m = \sum_{t \in m} r_t^2$ (the days t of month m); we model $\log RV_m$
- **Shuffle test**: permute the days at random and re-estimate d
 - ▶ the permutation keeps the distribution of the values and destroys their order: real memory gives $\hat{d} \approx 0$ after shuffling
- Why: memory in $|r_t|$ means that calm and turbulent periods last (volatility clustering)
 - ▶ it concerns the size of the moves, not their sign: the direction stays unpredictable

Prerequisites for Today (18/22): the shuffle test in a chart



Simulated: ACF of $|r_t|$ positive up to lag 200, LW $\hat{d} = 0.28$; after shuffling, inside the band, $\hat{d} = -0.04$

Prerequisites for Today (19/22): GARCH and FIGARCH weights



- GARCH(1,1): $\alpha\beta^{k-1}$, $\alpha = 0.08$, $\beta = 0.9$
- FIGARCH: $\lambda_k = -\pi_k$, $d = 0.4$
- - - $dk^{-1-d}/\Gamma(1-d)$

- **GARCH(1,1)** (Chapter 5): $\sigma_t^2 = \omega + \alpha\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2$
 - ▶ σ_t^2 : the conditional variance of day t ; ε_t : the shock; $\omega > 0$, $\alpha, \beta \geq 0$: the parameters
 - ▶ substituting σ_{t-1}^2 , σ_{t-2}^2 , ... again and again (ARCH(∞) form):

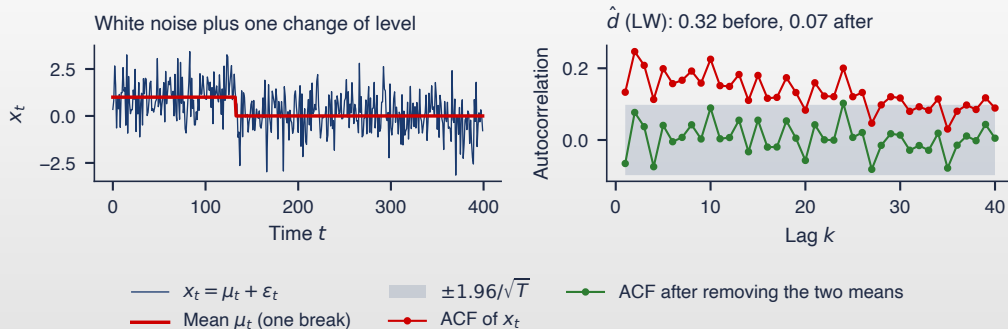
$$\sigma_t^2 = \frac{\omega}{1-\beta} + \alpha \sum_{k \geq 1} \beta^{k-1} \varepsilon_{t-k}^2$$

- ▶ weight of ε_{t-k}^2 : $\alpha\beta^{k-1}$, exponential
- **FIGARCH(0, d , 0)** Baillie, Bollerslev and Mikkelsen (1996): $\sigma_t^2 = \omega + [1 - (1-L)^d] \varepsilon_t^2$
 - ▶ weights $\lambda_k = -\pi_k \approx dk^{-1-d}/\Gamma(1-d)$: hyperbolic
 - ▶ chart, $k = 100$: $2.4 \cdot 10^{-6}$ (GARCH) against $4.3 \cdot 10^{-4}$ (FIGARCH)

Prerequisites for Today (20/22): spurious long memory

- A short-memory series with a change of level: $x_t = \mu_t + \varepsilon_t$, where the mean μ_t changes once (a **break**)
 - ▶ the ACF stays positive at long lags and $\hat{d} > 0$, although there is no memory [Diebold and Inoue \(2001\)](#)
 - ▶ remedy: estimate the mean of each regime, subtract it and re-estimate d
- **Monte Carlo band**: simulate many i.i.d. series of the same length, estimate d on each
 - ▶ the 2.5% and 97.5% quantiles of these $\hat{d}$: the 95% band of $\hat{d}$ when there is no memory; i.i.d.: independent and identically distributed
- An estimate $\hat{d}$ outside the band: evidence of memory, or of a break; inside: compatible with no memory

Prerequisites for Today (21/22): a break that looks like long memory



- Left: white noise whose mean falls by 1 after one third of the sample ($T = 400$)
- Right: the ACF stays above the band for 40 lags and LW gives $\hat{d} = 0.32$; after removing the two means, $\hat{d} = 0.07$

Prerequisites for Today (22/22): monthly and 12-month inflation rates

- P_t : the price index of month t ; **monthly rate** $\pi_t^m = 100 \Delta \ln P_t = 100(\ln P_t - \ln P_{t-1})$
 - ▶ seasonal adjustment by monthly means: subtract from each month (January, February, ...) its average over all years
- The **12-month rate** is the sum of the last 12 monthly rates

$$100(\ln P_t - \ln P_{t-12}) = \sum_{j=0}^{11} \pi_{t-j}^m$$

- ▶ two consecutive 12-month rates share 11 monthly rates (overlapping sums)
- ▶ even for i.i.d. monthly rates, the 12-month rate has $\rho(k) = (12 - k)/12$ for $k < 12$: a strong autocorrelation built by the definition

A1: fractional weights and the ACF [Solved]

Task

- A series is ARFIMA(0, 0.25, 0) with $\sigma^2 = 1$.
- 1. Compute $\pi_1, \dots, \pi_4$ of $(1 - L)^{0.25}$.
- 2. Compute $\rho(1)$ and $\rho(2)$, and give H and the class of the process.
- 3. Compare $\rho(5)$ with that of an AR(1) with the same $\rho(1)$.
- Report: four weights, two autocorrelations, H , two values of $\rho(5)$.

Solution

- 1. $\pi_1 = -0.25$; $\pi_2 = -0.25 \cdot 0.75/2 = -0.0938$;
 $\pi_3 = -0.0938 \cdot 1.75/3 = -0.0547$;
 $\pi_4 = -0.0547 \cdot 2.75/4 = -0.0376$
- 2. $\rho(1) = 0.25/0.75 = 0.333$;
 $\rho(2) = 0.333 \cdot 1.25/1.75 = 0.238$; $H = 0.75$: stationary, long memory
- 3. ARFIMA: $\rho(5) = 0.151$; AR(1): $0.333^5 = 0.0041$, about 37 times smaller

A2: which range is d in? [Proposed]

Solution discussed in class

Task

- ▣ Two series: (a) ARFIMA(0, 0.6, 0); (b) ARFIMA(0, -0.3, 0). Model: A1.
- ▣ 1. Compute π_1 and π_2 of $(1 - L)^d$ for each.
- ▣ 2. Say whether each series is stationary, invertible and mean-reverting.
- ▣ 3. For (b) compute $\rho(1)$ and $\rho(2)$; for (a) compute $\rho(1)$ of the first difference.
- ▣ Report: four weights, two classifications, three autocorrelations.

A3: R/S by hand and H from three points [Solved]

Task

- Block: $x = (0.6, 0.4, 0.5, 0.3, -0.4, -0.6, -0.2, -0.5)$. An R/S plot gives $(R/S)_n = 4.1; 9.6; 22.4$ for $n = 16, 64, 256$.
- 1. Compute $\bar{x}$, the cumulative deviations Y_k , R , S and R/S .
- 2. Compute H as the slope of $\log_{10}(R/S)_n$ on $\log_{10} n$, and $d = H - 0.5$.
- Report: R/S , H , d and one sentence.

Solution

- 1. $\bar{x} = 0.0125$; Y_k : 0.5875, 0.9750, 1.4625, 1.7500, 1.3375, 0.7250, 0.5125, 0; $R = 1.750$, $S = 0.457$, $R/S = 3.83$
- the signs are grouped (four positive, then four negative): the range is large, as for a persistent series
- 2. $\log_{10} n$: 1.204, 1.806, 2.408; $\log_{10}(R/S)$: 0.613, 0.982, 1.350; slope $H = 0.61$, $d = 0.11$: moderate persistence

A4: a GPH slope by hand [Proposed]

Solution discussed in class

Task

- ▣ For $m = 4$ frequencies: $X_j = 0.5; 1.6; 2.3; 3.0$ and $\log I(\lambda_j) = 1.2; 1.8; 2.0; 2.5$. Model: A3.
- ▣ 1. Compute $\bar{X}$, $\bar{y}$, $\sum(X_j - \bar{X})(y_j - \bar{y})$ and $\sum(X_j - \bar{X})^2$.
- ▣ 2. Compute $\hat{d}$ and its standard error ([Prerequisites for Today \(14/22\)](#)).
- ▣ 3. Test $H_0: d = 0$ at 5% and comment on the size of m .
- ▣ Report: four sums, $\hat{d}$, its SE, a decision and one sentence.

A5: a one-step ARFIMA forecast [Solved]

Task

- ▣ ARFIMA(0, 0.4, 0) with $\mu = 2$; the last four values: $x_T = 3.0$, $x_{T-1} = 2.5$, $x_{T-2} = 2.8$, $x_{T-3} = 1.5$.
- ▣ 1. Write the first four weights $-\pi_k$ of $(1 - L)^{0.4}$.
- ▣ 2. Compute $\hat{x}_{T+1}$ using these four lags.
- ▣ 3. Compare with an AR(1) with the same $\rho(1)$.
- ▣ Report: four weights, two forecasts and one sentence.

Solution

- ▣ 1. $-\pi_k$: 0.4; 0.12; 0.064; 0.0416
- ▣ 2. $\hat{x}_{T+1} =$
 $2 + 0.4(1.0) + 0.12(0.5) + 0.064(0.8) + 0.0416(-0.5) =$
 $2 + 0.4000 + 0.0600 + 0.0512 - 0.0208 = 2.490$
- ▣ 3. AR(1): $\phi = \rho(1) = 0.4/0.6 = 0.667$,
 $\hat{x}_{T+1} = 2 + 0.667 \cdot 1.0 = 2.667$; the AR(1) looks only at x_T , ARFIMA spreads the weight over the past (in practice over all T values)

A6: how long does a volatility shock count? [Proposed]

Solution discussed in class

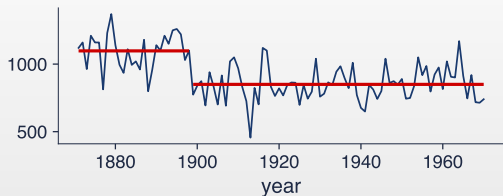
Task

- ▣ GARCH(1,1) with $\alpha = 0.10$, $\beta = 0.88$; FIGARCH(0, d , 0) with $d = 0.4$ (weights $\lambda_k = -\pi_k$). Model: A5.
- ▣ 1. Compute the weight of ε_{t-k}^2 in σ_t^2 for $k = 1, 2, 3$ in both models.
- ▣ 2. Compute both weights at $k = 20$ and $k = 100$; for FIGARCH use the approximation of λ_k in [Prerequisites for Today \(19/22\)](#), with $\Gamma(0.6) = 1.489$.
- ▣ 3. Explain which model keeps a crisis in today's volatility longer.
- ▣ Report: ten weights and one sentence.

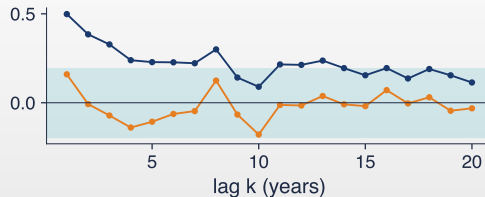
B1: does the Nile have long memory? [Solved]

- **Question:** is the persistence of the Nile flow long memory, or the effect of the change in 1898?
- **Inputs:** yearly flow at Aswan, 1871–1970 ($T = 100$)
- **Tasks**
 1. Plot the series and its ACF up to lag 20.
 2. Estimate H by R/S, d by GPH and local Whittle ($m = \lfloor T^{0.65} \rfloor$) and by exact ML of ARFIMA(0, d , 0).
 3. Subtract the mean of 1871–1898 and the mean of 1899–1970 and repeat step 2.
 4. Interpretation: does the Nile have long memory?
- **Report:** two ACF values, four estimates before and after, one sentence
- **Notebook:** section [B1](#)

B1: solution [Solved]



— Nile flow at Aswan (10^8 m^3)
 — regime means, break after 1898
 — ACF, raw series



— ACF, regime means removed
 95% band for white noise

- Raw: $\hat{\rho}(1) = 0.50$, $\hat{\rho}(10) = 0.09$; $H_{R/S} = 0.78$; GPH 0.44 (SE 0.15), LW 0.40 (SE 0.11), ML 0.36 (SE 0.07); $m = 19$
- Means 1098 and 850; after removing them: $\hat{\rho}(1) = 0.16$; $H_{R/S} = 0.63$; GPH -0.07 , LW -0.17 , ML 0.05 (SE 0.10)
- Interpretation: for 1871–1970 the evidence of long memory comes from one break; R/S stays above 0.5 because it is biased upwards in short samples

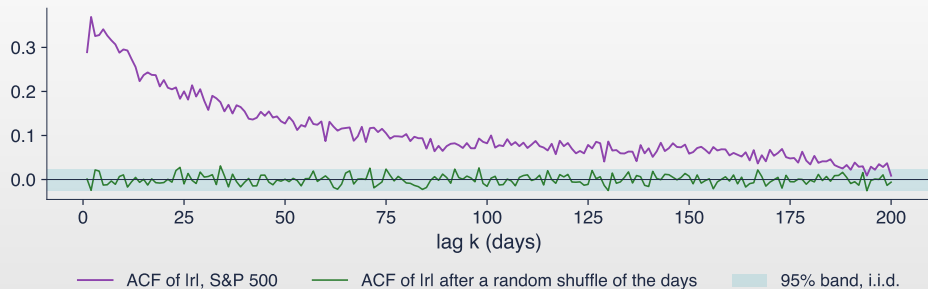
B2: long memory in Romanian inflation [Proposed]

- **Question:** how persistent is Romanian inflation, and does the way we measure inflation change the answer?
- **Inputs:** monthly HICP, 2005–2026: the monthly rate seasonally adjusted with the monthly means, and the 12-month rate; model: [B1](#)
- **Tasks**
 1. Estimate d of the monthly rate by GPH and local Whittle with $m = \lfloor T^a \rfloor$, $a = 0.5; 0.65; 0.8$.
 2. Fit ARFIMA(0, d , 0) and ARMA(1,1) by exact ML and compare their BIC.
 3. Estimate d of the 12-month rate by local Whittle and compare the two ACFs.
 4. Interpretation: why does the 12-month rate give d close to 1?
- **Report:** six estimates, two BIC values, one d , two sentences
- **Notebook:** section [B2](#)

B3: memory in S&P 500 volatility [Solved]

- **Question:** where is the memory of stock returns: in the returns or in their size?
- **Inputs:** daily log returns of the S&P 500 since 2000 and their monthly realised volatility
- **Tasks**
 1. Estimate d of r_t and of $|r_t|$ by local Whittle and GPH.
 2. Shuffle the days at random and re-estimate d of $|r_t|$; plot both ACFs of $|r_t|$.
 3. Estimate d of the monthly log realised volatility.
 4. Interpretation: what does the shuffle test show?
- **Report:** eight estimates, the chart and one sentence
- **Notebook:** section [B3](#)

B3: solution [Solved]



- $T = 6714$ days; r_t : LW -0.01 , GPH 0.02 (SE 0.03 , 0.04); $|r_t|$: LW 0.53 , GPH 0.56
- Shuffled $|r_t|$: LW -0.03 , GPH -0.03 ; ACF of $|r_t|$: 0.289 (lag 1), 0.127 (lag 50), 0.009 (lag 200); log RV (321 months): LW 0.41 , GPH 0.37
- Interpretation: the shuffle keeps every daily value and destroys their order; the memory disappears, so it lives in the clustering of calm and turbulent periods, not in the distribution of returns

B4: BET and EUR/RON [Proposed]

- **Question:** do the BET and the EUR/RON rate show the same memory, and is it stable before and after the 2008 crisis?
- **Inputs:** daily log returns of the BET (since 2000) and of the EUR/RON reference rate (since July 2005); model: [B3](#)
- **Tasks**
 1. Estimate d of r_t and $|r_t|$ by local Whittle on the full sample.
 2. Repeat on the subsamples up to 2007 and 2010–2026.
 3. Compare the estimates for r_t with the 95% Monte Carlo band of i.i.d. series of 1900 days.
 4. Interpretation: is the full-sample memory of BET returns real?
- **Report:** twelve estimates, one band and two sentences
- **Notebook:** section [B4](#)

C1: US inflation and monetary regimes [Proposed]

- **Question:** is the long memory of US inflation a property of inflation, or of the changes in monetary policy?
- **Inputs:** US monthly CPI inflation, annualised, 1947–2026; regimes 1947–1984, 1985–2019, 2020–2026; models: [B1](#), [B2](#)
- **Tasks**
 1. Estimate d by local Whittle on the full sample and in each regime.
 2. Remove the three regime means and re-estimate d .
 3. Propose one more check (a break test, a Markov-switching model, a forecast comparison).
 4. Interpretation: can a constant d describe 80 years of inflation?
- **Report:** five estimates, one check and a project plan
- **Notebook:** section [C1](#)

C2: audit an AI answer [Proposed]

- ▣ A student asked an AI assistant about the long memory of Romanian inflation and of the S&P 500. The answer:
- ▣ (a) Monthly inflation has $d = 0.31$, so its Hurst exponent is $H = d - 0.5 < 0$: it is anti-persistent.
- ▣ (b) The 12-month rate gives $d = 1.00$, so Romanian inflation has a unit root.
- ▣ (c) With $d = 0.31$ the series is stationary and its shocks die out.
- ▣ (d) Use `statsmodels.tsa.arima.model.ARIMA(x, order=(0, 0.31, 0))` to fit the ARFIMA model.
- ▣ (e) S&P 500 returns have $d = -0.01$ and $|r|$ has $d = 0.53$, so returns are predictable.
- ▣ (f) A significant GPH estimate proves long memory.
- ▣ **Tasks**
 1. For each statement, say whether it is correct; if not, give the correct statement and, where possible, the correct number from the notebook (section C2).
- ▣ **Report:** a list of six verdicts with one line of justification each

Key takeaways

- ▣ $(1 - L)^d$: weights by the recursion $\pi_k = \pi_{k-1}(k - 1 - d)/k$; ARFIMA(0, d , 0): $\rho(1) = d/(1 - d)$
- ▣ $H = d + 1/2$; R/S and GPH are slopes of log-log regressions
- ▣ Romanian monthly inflation: $d \approx 0.3$; S&P 500: memory in $|r_t|$, not in r_t
- ▣ Breaks can create long memory: the Nile, BET returns, US inflation
- ▣ An AI answer is a draft: check H against d , the library, and the data transformation

After the seminar

- Lecture 8 develops each topic of today
 - ▶ hyperbolic decay, fractional differencing, ARFIMA, the estimators, forecasting, volatility, spurious memory
- Try the [Proposed] tasks in the notebook
- C1 can grow into a team project: inflation memory and monetary regimes in Romania
- Reading: Baillie (1996); Graves et al. (2017); Ding, Granger and Engle (1993)

References (1/2)

- ▣ Baillie, R. T. (1996). Long memory processes and fractional integration in econometrics. *Journal of Econometrics*, 73(1), 5–59. doi:10.1016/0304-4076(95)01732-1
- ▣ Baillie, R. T., Bollerslev, T., & Mikkelsen, H. O. (1996). Fractionally integrated generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 74(1), 3–30. doi:10.1016/S0304-4076(95)01749-6
- ▣ Diebold, F. X., & Inoue, A. (2001). Long memory and regime switching. *Journal of Econometrics*, 105(1), 131–159. doi:10.1016/S0304-4076(01)00073-2
- ▣ Ding, Z., Granger, C. W. J., & Engle, R. F. (1993). A long memory property of stock market returns and a new model. *Journal of Empirical Finance*, 1(1), 83–106. doi:10.1016/0927-5398(93)90006-D
- ▣ Geweke, J., & Porter-Hudak, S. (1983). The estimation and application of long memory time series models. *Journal of Time Series Analysis*, 4(4), 221–238. doi:10.1111/j.1467-9892.1983.tb00371.x
- ▣ Granger, C. W. J., & Joyeux, R. (1980). An introduction to long-memory time series models and fractional differencing. *Journal of Time Series Analysis*, 1(1), 15–29. doi:10.1111/j.1467-9892.1980.tb00297.x

References (2/2)

- ▣ Graves, T., Gramacy, R., Watkins, N., & Franzke, C. (2017). A brief history of long memory: Hurst, Mandelbrot and the road to ARFIMA, 1951–1980. *Entropy*, 19(9), 437. doi:10.3390/e19090437
- ▣ Hosking, J. R. M. (1981). Fractional differencing. *Biometrika*, 68(1), 165–176. doi:10.1093/biomet/68.1.165
- ▣ Hurst, H. E. (1951). Long-term storage capacity of reservoirs. *Transactions of the American Society of Civil Engineers*, 116(1), 770–799. doi:10.1061/TACEAT.0006518
- ▣ Peng, C.-K., Buldyrev, S. V., Havlin, S., Simons, M., Stanley, H. E., & Goldberger, A. L. (1994). Mosaic organization of DNA nucleotides. *Physical Review E*, 49(2), 1685–1689. doi:10.1103/PhysRevE.49.1685
- ▣ Robinson, P. M. (1995). Gaussian semiparametric estimation of long range dependence. *The Annals of Statistics*, 23(5), 1630–1661. doi:10.1214/aos/1176324317
- ▣ Sowell, F. (1992). Maximum likelihood estimation of stationary univariate fractionally integrated time series models. *Journal of Econometrics*, 53(1–3), 165–188. doi:10.1016/0304-4076(92)90084-5